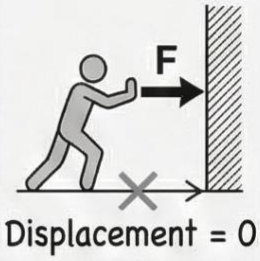
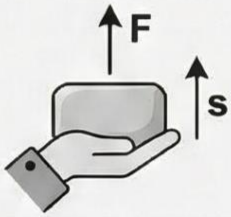
WORK AND ENERGY

1. Introduction to Work

- In daily life, 'work' means effort or tiredness.
- In Physics, work has a specific scientific meaning.
- **Example 1:** Pushing a wall — Force applied but no displacement → No work done.



- **Example 2:** Lifting a stone — Force + Displacement → Work done.

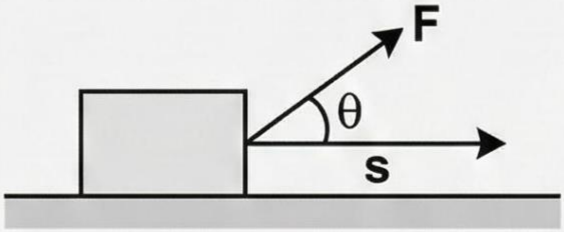


- **Conclusion:** Effort alone does not mean work in Physics.

2. Conditions for Work Done

- Three conditions must be satisfied:
  1) **Force** must act on the object.
  2) **Displacement** must occur.
  3) **Displacement** must be in the **direction** of force.
- Mathematical Expression:
  $W = F \times s \times \cos\theta$

$$\boxed{W = F\,s\cos\theta}$$

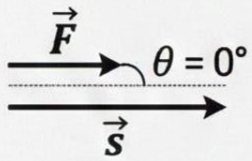


Important Note: If any one condition is not satisfied, work done = 0.

WORK AND ENERGY (Continued)

3. Three Important Cases of Work

Formula: $W = F\,s\cos\theta$

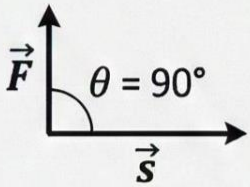


Case: $\theta = 0°$

$\cos 0° = 1$

→ Maximum Positive Work

Concept Note:
Force and displacement in same direction

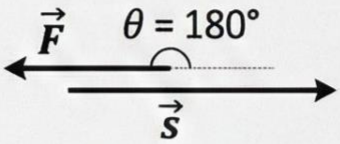


Case: $\theta = 90°$

$\cos 90° = 0$

→ Zero Work

Concept Note:
Example: Centripetal Force in circular motion



Case: $\theta = 180°$

$\cos 180° = -1$

→ Negative Work

Concept Note:
Example: Friction, Braking force

4. Energy – Concept and Definition

Energy = Ability to do work

- Energy can exist even without motion.
- Energy is a scalar quantity.
- SI Unit: Joule (J)
- CGS Unit: erg
- <u>Work = Energy Transfer</u>

5. Types of Energy (Overview)

- Kinetic Energy
- Potential Energy
- Heat Energy
- Electrical Energy
- Solar Energy
- Chemical Energy
- Nuclear Energy

6. Kinetic Energy (KE)

Energy possessed by a body due to its motion.

$$KE = \frac{1}{2}mv^2$$

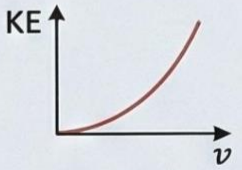
Note: Velocity is more influential than mass.

Concept Points

- Kinetic energy depends on both mass and velocity.
- Velocity is squared (v^2) → small increase in velocity causes large increase in KE.
- Doubling velocity increases KE four times.

Concept Example

Car comparison: A light fast car may have more KE than a heavy slow truck.



KE vs v curve

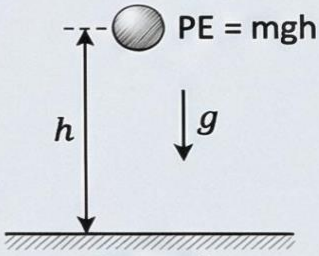
7. Potential Energy (PE)

Energy possessed by a body due to its position or configuration.

$$PE = mgh$$

Concept Points

- Depends on mass, gravitational acceleration, and height.
- Reference level must be defined.
- Potential energy can be negative depending on reference level.



Ground
(Reference Level, h=0)

MCQ Trap:
Important: Potential Energy is relative, not absolute.

8. Work–Energy Theorem

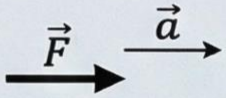
Net Work Done = Change in Kinetic Energy (ΔKE)

$$W_{net} = KE_{final} - KE_{initial}$$

ΔKE

Concept Points

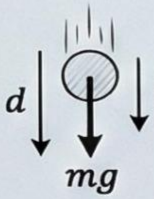
- Positive net work → KE increases.
- Negative net work → KE decreases.
- Zero net work → Constant velocity motion.



Force doing work

Examples

- Falling object (gravity does positive work).
- Accelerating car (engine does positive work).
- Object moving at constant speed (net work = 0).



Gravity does +W

9. Law of Conservation of Energy

"Energy cannot be created or destroyed; it can only transform from one form to another."

- Total energy of an isolated system remains constant.
- Energy changes form but total amount remains conserved.

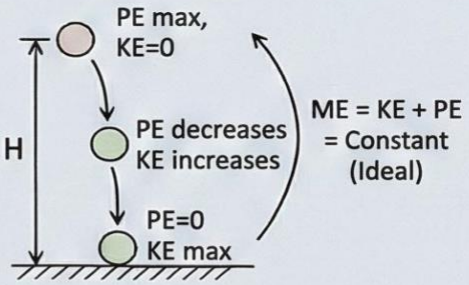
Common Misconception:
"Energy is lost" → Actually, mechanical energy converts into heat, sound, etc.

10. Mechanical Energy

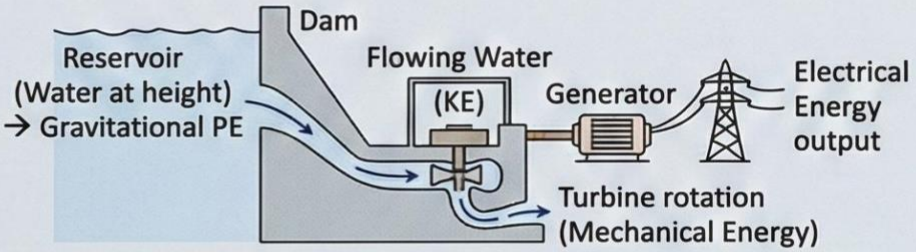
Mechanical Energy = KE + PE

- **Conservation Condition:**
  - No friction
  - No air resistance
  - No non-conservative forces
- **Concept Points**
  - In ideal systems, mechanical energy remains constant.
  - In real-world systems, mechanical energy decreases due to friction.
  - However, total energy remains conserved.

Ideal System vs Real System	
Ideal: ME = constant Ball falling KE increases PE decreases Sum constant.	**Real: ME decreases** Friction/Air resistance KE+PE < Initial ME Some energy becomes Heat/Sound.



11. Energy Transformation in Real Life



Gravitational PE converts to KE, then mechanical energy, then electrical energy.

- **Everyday Examples**

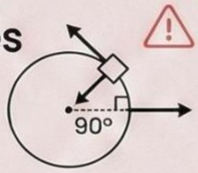
Tube Light: Electrical → Light + Heat
Solar Panel: Solar → Electrical
Electric Fan: Electrical → Mechanical
Battery: Chemical → Electrical

Heater: Electrical → Heat
Torch: Chemical → Electrical → Light
Electromagnet: Electrical → Magnetic
Photosynthesis: Solar → Chemical

CONCEPTUAL TRAPS & EXAM REVISION

12. Common Conceptual Traps (Prelims Focus)

Centripetal force does zero work (Force ⊥ Displacement).



Force without displacement = Zero Work.

Potential Energy depends on reference level.

Statement "Energy is destroyed" is incorrect.

Mechanical Energy ≠ Total Energy.

13. Competitive Exam Application

- Questions are mostly concept-based MCQs.
- MPSC / UPSC Prelims focus on logical understanding.
- Avoid blind formula memorization.
- Understand angle concept and energy transformation logic.

Logic-driven approach > Formula memorization

14. Summary & Key Takeaways

- ✓ Work = Energy Transfer
- ✓ KE depends strongly on velocity (v^2 factor).
- ✓ PE depends on reference level.
- ✓ Work changes Kinetic Energy (Work–Energy Theorem).
- ✓ Total Energy is always conserved.